

NATIONAL COLLEGE OF IRELAND: SCHOOL OF COMPUTING

Master of Science in Data Analytics (MSCDAD) | Research Practicum WEEKLY PROGRESS REPORT: WEEK 1

Student Name:	Rohitkumar Amrittal Jaiswal (ID: x25119613)
Supervisor / Lecturer:	Dr. Thanos Staikopoulos
Project Title:	Coupling Port Congestion & Customs Delay Prediction with Carbon-Aware Multi-Objective Route Optimisation
Reporting Period / Date:	Week 1 (Kickoff, Technical Initialization & Baseline Benchmark) Tuesday, 29 September 2026

1. Activities Completed This Week (What?)

- Repository Architecture & Packaging:** Initialized the formal Git repository with modern PEP 621 packaging (irishlogix v0.1.0) and automated testing harness.
- Automated CSO Data Pipeline (cso_loader.py):** Integrated the Central Statistics Office (CSO) PxStat API; successfully ingested and parsed 444 vessel arrivals (TBQ01), 2,331 Ro-Ro freight records (TBQ04), and 3,600 trade region records (TBQ05). Created unified port congestion matrix.
- Jupyter Notebook Development:** Created and executed notebooks/01_week1_data_ingestion_and_exploration.ipynb with live data loading, longitudinal trend visualisations (2017 to 2026), and baseline metrics.
- Irish Freight Corridor Graph (corridor_network.py):** Implemented NetworkX road network graph with 11 strategic nodes (Dublin, Cork, Rosslare, M50, Athlone, Limerick, Galway, Waterford, Dundalk) and 15 bidirectional corridors with tolls, speed limits, and distance matrices.
- Baseline FIFO Dispatcher Simulator (baseline_dispatcher.py):** Coded and executed empirical baseline simulation across 50 benchmark consignments: 8,636.5 km distance, 104.4 driving hrs, **63.5 idle waiting hrs** (averaging **1.27 hrs wasted idling per trip**), 2,879.3 L diesel, 7,601.3 kg CO2e, and €14,791.21 total cost (avg €295.82/trip).
- Unit Test Suite:** Authored 9 unit tests across data ingestion, graph routing, and simulation; 100% passing (pytest tests -v in 1.10s).

2. Primary Public Dataset Sources & URLs

Dataset Name	URL	Description
CSO TBQ01 (Vessel Arrivals)	https://data.cso.ie/table/TBQ01	Quarterly port ship arrivals for C1 classifier
CSO TBQ04 (Ro-Ro Units)	https://data.cso.ie/table/TBQ04	Trailer counts (inward/outward) by port
CSO TBQ05 (Trade Regions)	https://data.cso.ie/table/TBQ05	Great Britain vs EU freight flows (post-Brexit)
EMODnet Vessel Density	https://emodnet.ec.europa.eu/en/human-activities	AIS coastal marine traffic density grids
OpenStreetMap Ireland	https://download.geofabrik.de/europe/ireland-and-northern-ireland.html	National freight motorway and primary corridors

3. Evaluation & Critical Reflection (So What?)

- Validation of Research Gap:** The Week 1 baseline simulation empirically validates the core research rationale: without predictive delay awareness, freight hauliers waste an average of 1.27 hours per shipment waiting at terminal gates and customs inspections, creating direct financial and carbon penalties.
- Data Grounding:** Live CSO PxStat API data guarantees verified, open historical maritime records (2017 to 2026) for Dublin, Cork, and Rosslare, de-risking Component 1 model development.

4. Action Plan & Next Steps (Now What?)

- Oral Pitch (1 October):** Deliver the timed 1-2 minute project pitch adhering to Problem -> Gap -> Idea -> Method -> Contribution.
- Supervisory Alignment:** Present Week 1 code, Jupyter notebook, and benchmark findings to Dr. Thanos Staikopoulos during scheduled supervisory review.
- EMODnet & AIS Ingestion:** Incorporate EMODnet vessel density metrics into the maritime congestion feature store.

5. Supervisor Meeting Log & Agreed Actions

Agenda Items:	1. CA2 Proposal & Ethics sign-off. 2. 1 October Pitch rehearsal. 3. Demonstration of Week 1 technical deliverables: Git repo, CSO pipeline, Jupyter notebook, baseline FIFO simulation.
Supervisor Guidance:	- Emphasize empirical gap between unbuffered spreadsheet planning and predictive optimization. - Ensure baseline simulation parameters are grounded in Irish logistics economics.
Agreed Actions:	1. Deliver 1-2 minute pitch on 1 October. 2. Submit weekly progress updates every Tuesday. 3. Begin feature engineering on ingested CSO port traffic data.

6. Project Milestone & Gantt Tracking

Milestone ID	Target Week	Status	Details / Deliverables
M1	Week 3	ADVANCED / ON SCHEDULE	Baseline FIFO model operational; CSO data ingested; corridor graph built

M2	Week 9	PLANNED	XGBoost models (C1, C2) coupled with Pareto Optimizer (C3)
M3	Week 13	PLANNED	Dual-system comparative evaluation & Streamlit UI dashboard
M4	Week 15	PLANNED	Final MSc Dissertation, code repository, and Viva Voce

Student Signature: Rohitkumar Amritlal Jaiswal

Date: 29 September 2026

Supervisor Signature: _____

Date: _____